Exploratory Data Analysis (EDA) Report

Dataset: Sample Sales Orders | Tool: Python (Google Colab)

Overview

This analysis uses a small sample sales dataset to check data quality and understand order quantities, revenue, product performance and monthly sales patterns.

Data Cleaning

The original dataset contained 10 records. One missing customer age was filled using the median age (30). One duplicate order was removed. Data types and values were checked, and the order date was converted to a date format. An unusually large quantity of 100 was identified using the IQR method and excluded for this practice exercise. The final cleaned dataset contains 8 orders.

Basic Statistics

The 8 orders contain 21 items and generate total revenue of ₹2,37,500. Average revenue per order is ₹29,687.50, and the average quantity per order is 2.63 items. Product prices range from ₹500 to ₹50,000.

Correlation Analysis

A correlation matrix was calculated for quantity, price, customer age and revenue. This was used to examine how these numerical fields move together. Since the dataset contains only 8 cleaned orders, any correlation should be treated as an initial observation rather than a general business conclusion.

Trend Analysis

Monthly revenue was ₹1,02,500 in January, ₹16,500 in February, ₹1,02,000 in March and ₹16,500 in April 2026. January and March recorded the highest revenue in this sample. Laptop sales contributed ₹2,00,000, followed by monitors (₹24,000), keyboards (₹9,000) and mice (₹4,500).

Key Business Insights

Laptop sales account for most of the revenue, even though the dataset also includes lower-priced accessories. Monthly revenue fluctuates considerably across the four months. The small sample size limits how much we can conclude about longer-term business performance.

Conclusion

The dataset was cleaned and analyzed using missing-value treatment, duplicate removal, validation, outlier detection, descriptive statistics, correlation and trend analysis. The cleaned CSV is submitted separately. The results summarize this practice dataset and are not intended to represent a real business.